

Python for Data Engineering – 10 Week Roadmap

Week 1 – Python Fundamentals

Variables, data types, operators, strings, input/output. Resources: <https://docs.python.org/3/tutorial/> <https://www.w3schools.com/python/> <https://www.learnpython.org/> Project: Calculator + Temperature converter

Week 2 – Control Flow & Functions

if/else, loops, functions, lambda. Resources: <https://realpython.com/defining-your-own-python-function/> <https://docs.python.org/3/tutorial/controlflow.html> Project: Number guessing game

Week 3 – Data Structures

Lists, dicts, sets, comprehensions. Resources: <https://docs.python.org/3/tutorial/datastructures.html> <https://realpython.com/python-dicts/> Project: Contact book system

Week 4 – File Handling & Errors

CSV, JSON, file reading/writing, try/except. Resources: <https://realpython.com/read-write-files-python/> <https://realpython.com/python-json/> Project: CSV cleaner

Week 5 – Pandas Basics

DataFrames, filtering, sorting. Resources: <https://www.kaggle.com/learn/pandas> <https://pandas.pydata.org/docs/> Project: Sales data analysis

Week 6 – Pandas Advanced

Groupby, merge, pivot, datetime. Resources: <https://realpython.com/pandas-groupby/> https://pandas.pydata.org/docs/user_guide/ Project: Amazon dataset analysis

Week 7 – APIs & Databases

Requests, JSON APIs, PostgreSQL, SQLAlchemy. Resources: <https://realpython.com/api-integration-in-python/> <https://docs.sqlalchemy.org/> Project: Weather API → DB pipeline

Week 8 – Data Engineering Python

Logging, config, CLI, ETL scripts. Resources: <https://realpython.com/python-logging/> <https://realpython.com/python-yaml/> Project: Full ETL pipeline

Week 9–10 – Advanced + Projects

NumPy, Parquet, testing, validation. Resources: <https://numpy.org/doc/> <https://docs.pytest.org/>
Projects: API → DB ETL CSV → Parquet pipeline Real-time data collector